Section 2.7 — Function Algebra

Math 103: College Algebra

1 Combined functions

Define the following functions:

$$\begin{aligned} f &= \{(-7,-9),(-1,-3),(1,\ \ 3),(5,\ \ 0),(9,-8)\} \\ g &= \{(-7,-4),(\ \ 1,-8),(5,\ \ 1),(7,\ \ 0),(9,-7)\} \\ h &= \{(-7,\ \ 0),(-1,-5),(1,-7),(7,-1),(9,-5)\} \end{aligned}$$

Find each of the following. If the given value is not in the domain of the given function, write "undefined."

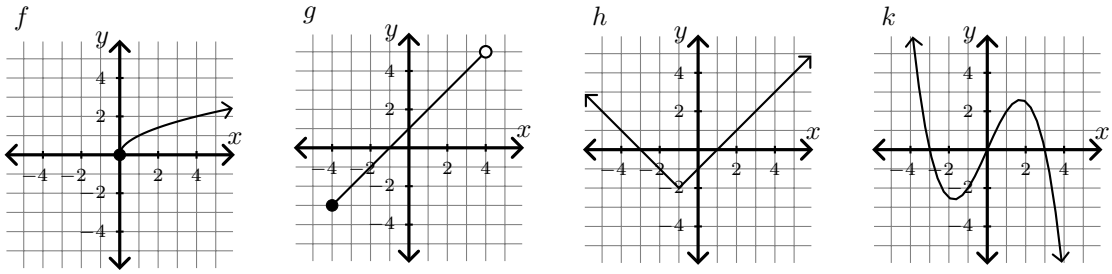
1. $(f \cdot g)(9)$
2. $(f - h)(5)$
3. $(h - g)(7)$
4. $(g \cdot h)(-7)$
5. $(f + g)(-7)$
6. $(g - f)(1)$
7. $\left(\frac{g}{h}\right)(-7)$
8. $\left(\frac{f}{g}\right)(5)$
9. $(h \cdot f)(7)$
10. $(g + h)(7)$
11. $\left(\frac{h}{f}\right)(-1)$
12. $(h + f)(-1)$

Find each domain.

13. $\operatorname{Dom} f$
14. $\operatorname{Dom} g$
15. $\operatorname{Dom} h$
16. $\operatorname{Dom}(g + h)$
17. $\operatorname{Dom}(h - f)$
18. $\operatorname{Dom}(f \cdot g)$
19. $\operatorname{Dom}\left(\frac{h}{g}\right)$

2 Combined Functions from Graphs

Consider the functions graphed below.



Find each of the following. If the given value is not in the domain of the given function, write "undefined."

20. $(f + g)(1)$
21. $(h - k)(3)$
22. $(g \cdot h)(-1)$
23. $\left(\frac{f}{h}\right)(0)$
24. $(h - g)(4)$
25. $(k + f)(-3)$
26. $(g \cdot k)(3)$
27. $\left(\frac{k}{h}\right)(-3)$

Find each domain.

28. $\operatorname{Dom} f$
29. $\operatorname{Dom} g$
30. $\operatorname{Dom} h$
31. $\operatorname{Dom} k$
32. $\operatorname{Dom}(f - g)$
33. $\operatorname{Dom}(k + f)$
34. $\operatorname{Dom}(h \cdot k)$
35. $\operatorname{Dom}\left(\frac{g}{k}\right)$

3 Combined Functions

$f(x) = 3x^2 - 8x + 4$ $\quad g(x) = x^3 + 7x^2 - 4x - 28$ $\quad h(x) = x - 2$ $\quad k(x) = 3x^2 - 19x - 14$

Use the functions above to find the following.

36. $(f + k)(2)$
37. $(g - h)(2)$
38. $(k \cdot g)(2)$
39. $\left(\frac{h}{f}\right)(2)$

Write expressions for each of the following. Simplify your answer as much as possible.

40. $(f - k)(x)$
41. $(g + f)(x)$
42. $(f \cdot h)(x)$
43. $(h - g)(x)$
44. $(h + k)(x)$
45. $(f \cdot k)(x)$

Write expressions for each of the following. Do not simplify your answer.

46. $\left(\frac{f}{h}\right)(x)$
47. $\left(\frac{g}{h}\right)(x)$

Find each domain.

48. $\operatorname{Dom} f$
49. $\operatorname{Dom} g$
50. $\operatorname{Dom} h$
51. $\operatorname{Dom} k$
52. $\operatorname{Dom}(f + g)$
53. $\operatorname{Dom}(h - k)$
54. $\operatorname{Dom}(f \cdot h)$
55. $\operatorname{Dom}\left(\frac{g}{k}\right)$